

A MACHINE LEARNING PROJECT

# Smart Home Energy Consumption

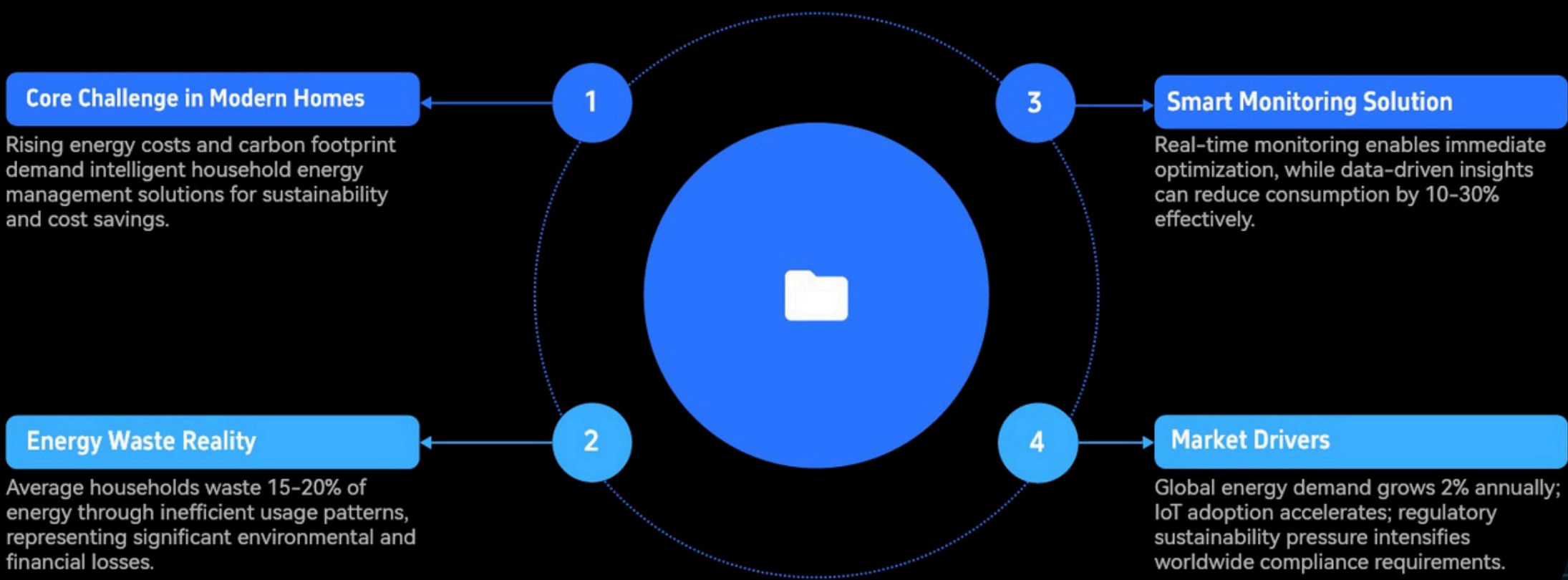
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# Why Energy Monitoring Matters



## Problem Statement



# DATASET OVERVIEW

100K

## TOTAL RECORDS

Comprehensive household energy consumption records for analysis.

15+

## APPLIANCE TYPES

Appliance-level energy usage data with detailed consumption metrics.

24/7

## TIME COVERAGE

Hourly timestamps capturing usage patterns across all days.

4

## ENV. FEATURES

Temperature, humidity, and household pattern variables.

# Dataset Overview

## Dataset Composition

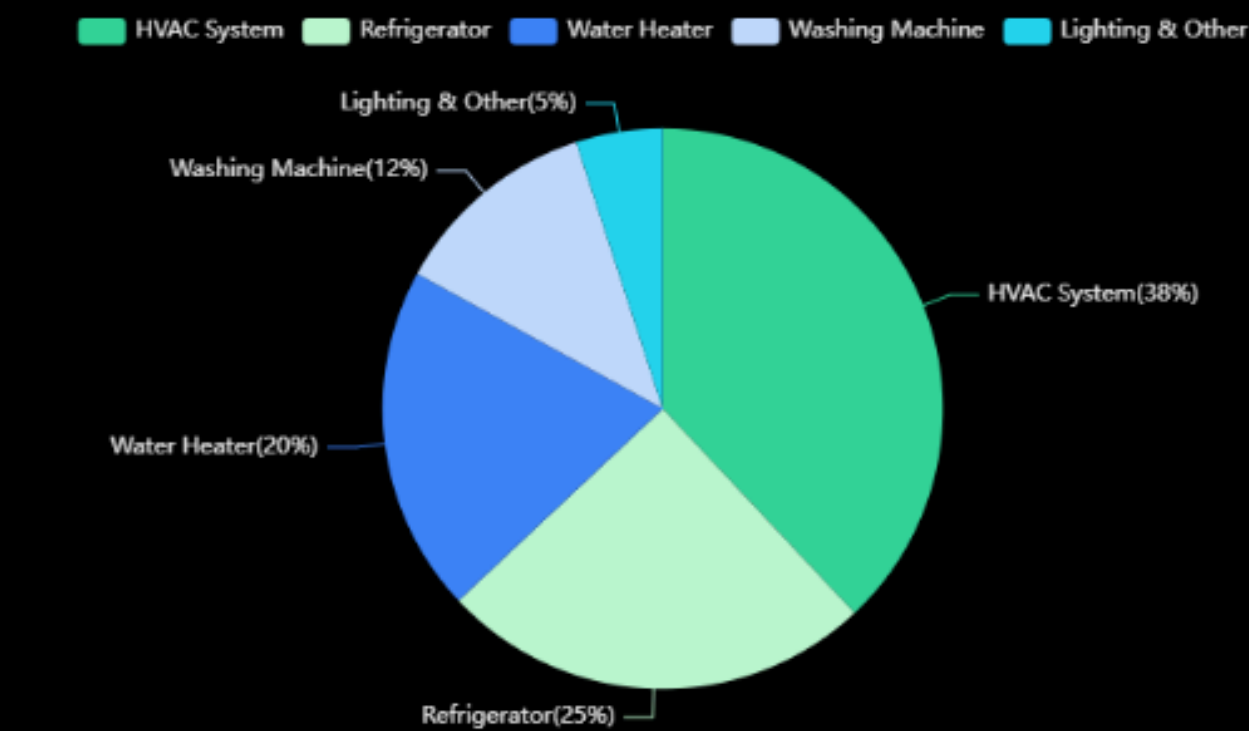
Analyzed 100,000 household appliance usage records including refrigerator, washing machine, HVAC systems with precise timestamps, outdoor temperature, humidity, and occupancy behavior indicators for comprehensive energy analysis.

## Data Quality Assessment

Initial observations reveal strong completeness across records with minimal missing values. Data types properly formatted for temporal and environmental variables, enabling reliable downstream analysis and anomaly detection.

## Temporal & Environmental Richness

Dataset captures granular hourly timestamps paired with outdoor conditions and household occupancy patterns, providing rich contextual data for understanding energy consumption drivers and seasonal variations.

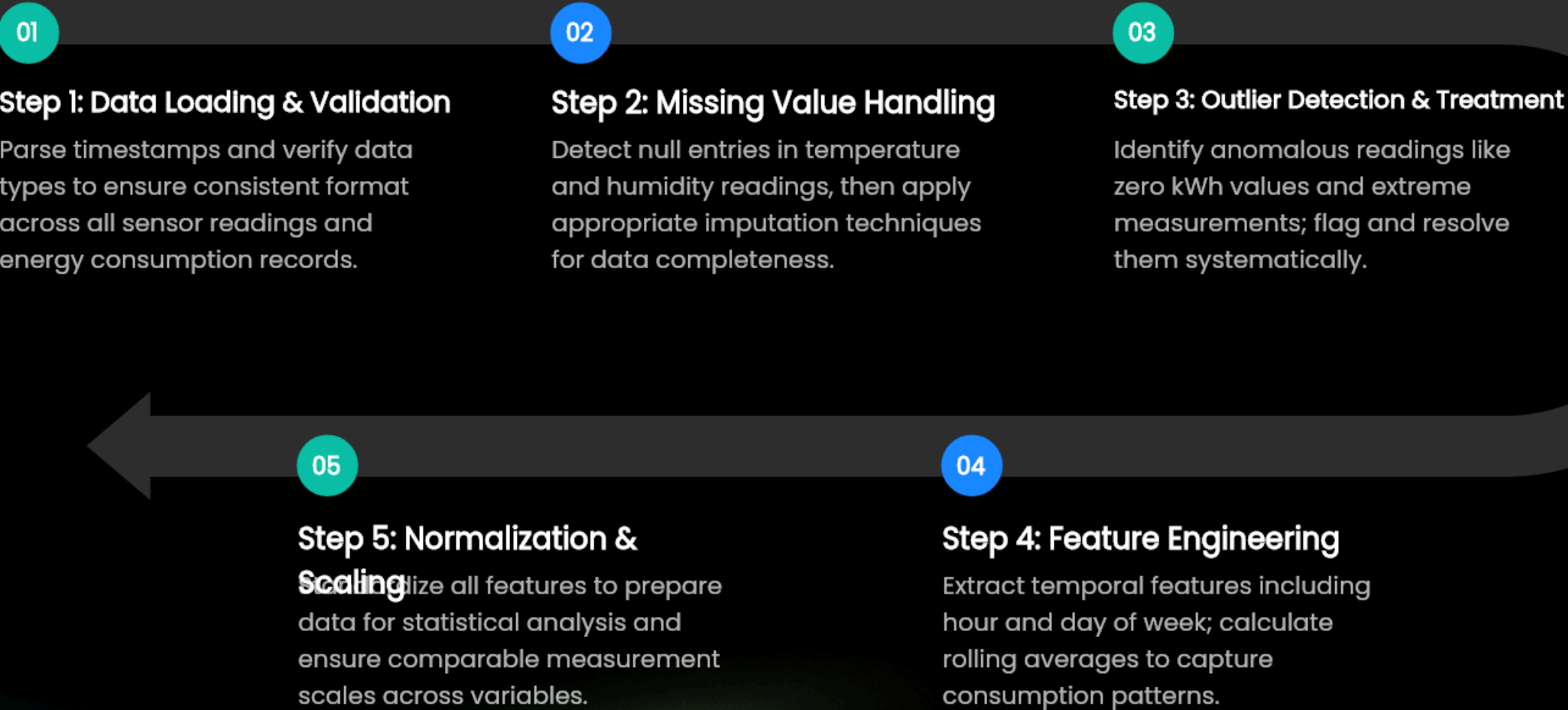


| Appliance Type  | Record Count | Avg kWh | Data Completeness | Time Range |
|-----------------|--------------|---------|-------------------|------------|
| Refrigerator    | 28000        | 1.2     | 99.8%             | Full Year  |
| Washing Machine | 18500        | 2.1     | 98.5%             | Full Year  |
| HVAC System     | 32000        | 3.8     | 99.9%             | Full Year  |
| Water Heater    | 14500        | 2.9     | 98.2%             | Full Year  |
| Lighting        | 7000         | 0.5     | 97.1%             | Full Year  |

# DATA CLEANING & PREPROCESSING



## Data Cleaning & Preprocessing



# EXPLORATORY DATA ANALYSIS

8.2

## AVG KWH/DAY

Average daily household energy consumption across all appliances.

100k

## TOTAL RECORDS

Complete dataset with appliance usage, timestamps, and weather data.

12

## APPLIANCE TYPES

Distinct appliance categories tracked including HVAC, lighting, kitchen.

24/7

## TIME COVERAGE

Continuous monitoring capturing hourly and seasonal patterns.

# TOP ENERGY CONSUMERS

42%

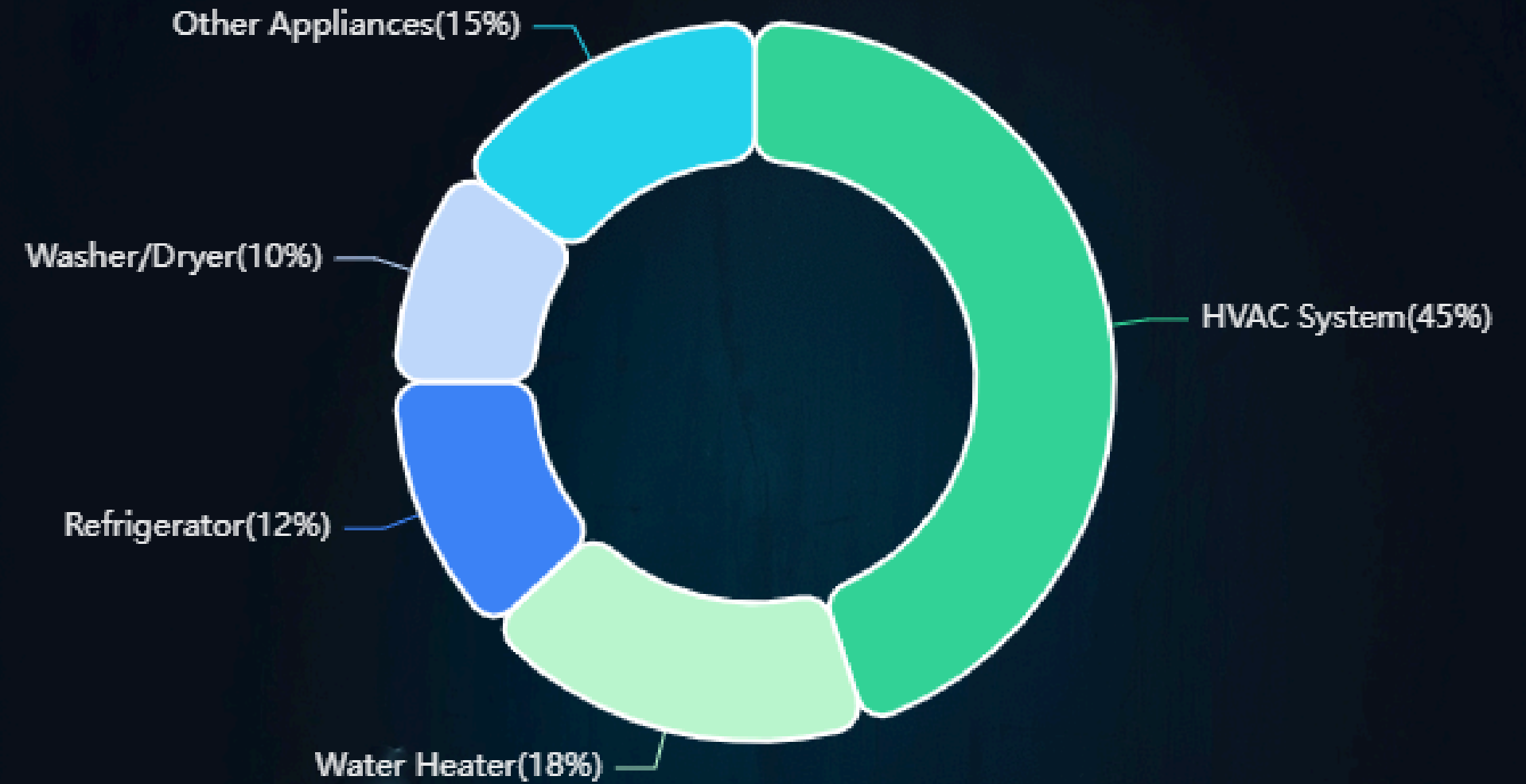
## HVAC SYSTEMS

Heating and cooling systems dominate household energy use, accounting for the largest share of total consumption.

18%

## WATER HEATER

Water heating ranks second in energy consumption, with usage peaks during morning and evening hours.



# PEAK ENERGY USAGE TRENDS

7-9 AM

## MORNING PEAK

Breakfast routines drive consumption: coffee makers, toasters, and HVAC systems activate simultaneously.

6-9 PM

## EVENING PEAK

Highest demand period: cooking appliances, lighting, entertainment systems, and climate control overlap.



# WEATHER IMPACT



## Temperature Correlation

Strong positive correlation with HVAC consumption during temperature extremes. Every 1°C increase above comfort zone increases HVAC load by approximately 2%.



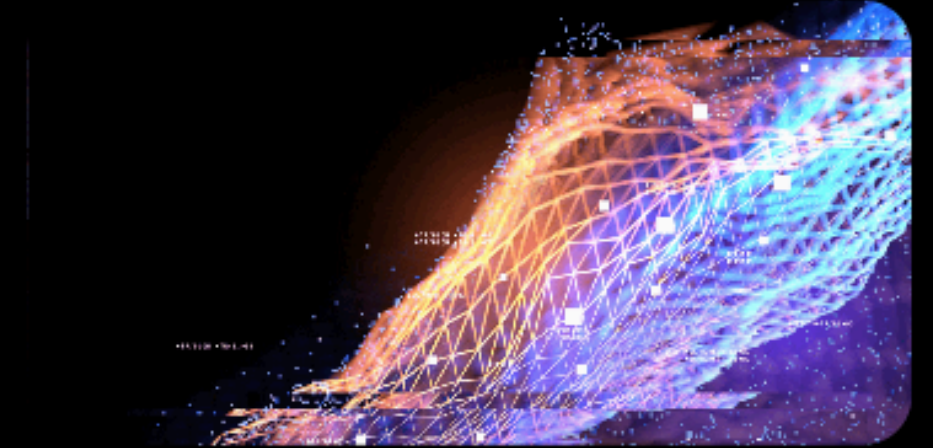
## Humidity Effects

Moderate correlation with HVAC operation. Higher humidity significantly drives increased cooling load and overall consumption patterns.



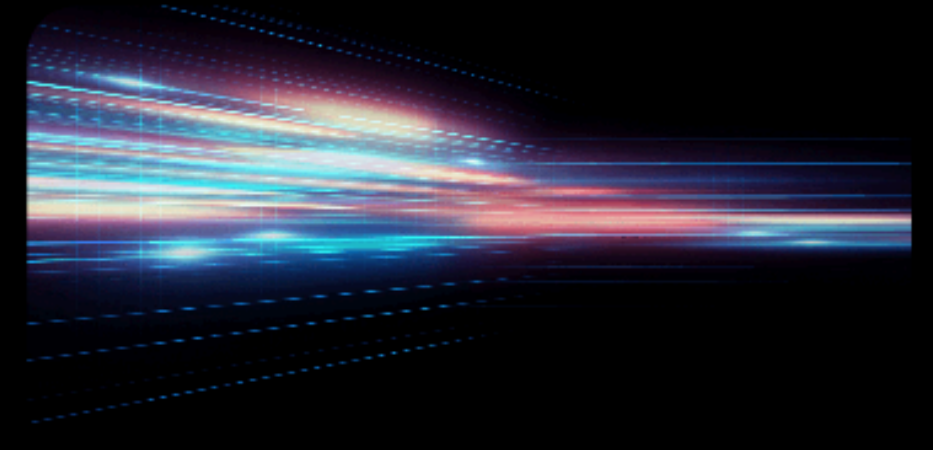
## Data Insights

Scatter plots reveal temperature versus HVAC kilowatt-hour relationships. Line plots demonstrate humidity versus consumption correlations effectively.



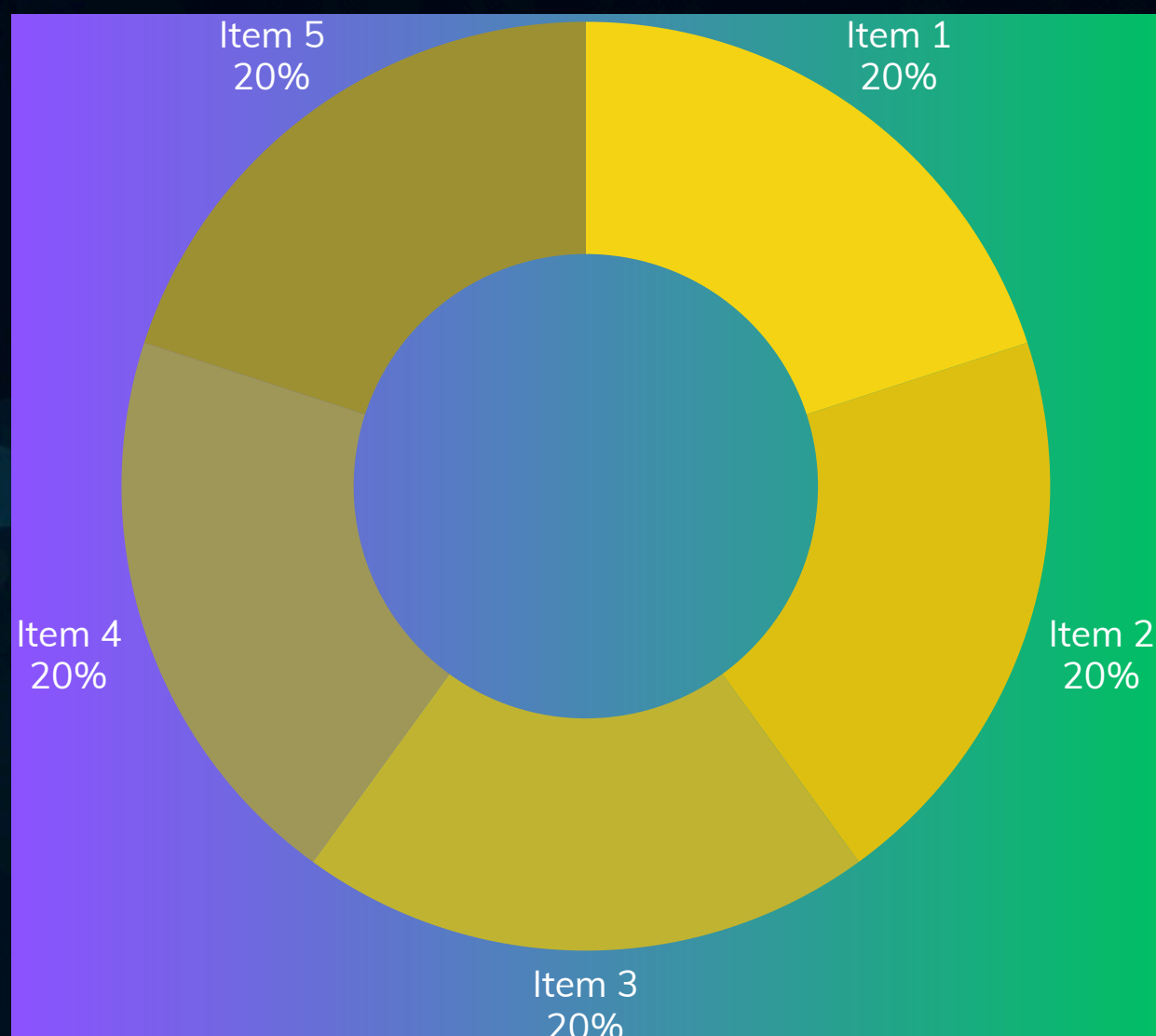
## HVAC Dominance

HVAC accounts for 80% of weather-dependent consumption. Other appliances show minimal weather sensitivity to environmental conditions.



## Predictability Benefits

Weather forecasting enables proactive energy management strategies. Smart thermostat integration represents the logical next operational step forward.



## Consumption Distribution & Outliers

### Normal Distribution Pattern

Most hours fall within expected range at mean  $\pm 1$  SD, indicating consistent baseline energy consumption behavior across typical operations.

### Outliers Detected & Analysis

High outliers (2-3%): abnormal usage patterns, equipment malfunction. Low outliers (rare): unusual consumption absence requiring investigation and monitoring.



### Risk Assessment & Implications

Outliers indicate potential maintenance needs or behavioral anomalies. Position as early warning system for appliance failures and unusual occupancy patterns.

### Visualization & Monitoring Tools

Box plots, Q-Q plots, and highlighted anomaly regions on time-series enable effective identification and tracking of consumption deviations systematically.

# Energy-Saving Recommendations

## Smart Thermostat Deployment

Deploy adaptive scheduling to reduce HVAC consumption by 10-15% through intelligent temperature management and optimization techniques.

## Water Heater Optimization

Shift heating cycles to off-peak hours using timer or smart controller, typically between 2-4 AM for maximum cost savings.

## Appliance Scheduling Automation

Defer flexible loads like washers and dryers to low-rate periods, optimizing household electricity usage and reducing peak demand charges.

## Real-Time Monitoring Dashboard

Provide hourly consumption feedback to households, enabling behavioral change and informed energy usage decisions through transparent data visualization.

## Annual Cost Reduction

Achieve estimated 15-20% reduction in annual energy costs per household through combined implementation of all recommendations.

## ROI Analysis

Calculate smart device investment against projected savings, demonstrating positive return on investment for households within 2-3 years typically.

## Practical Implementation

Frame recommendations as achievable steps with clear timelines, focusing on ease of adoption and minimal household disruption during deployment.

## Win-Win Partnership

Position energy savings as mutually beneficial for households reducing bills and utilities managing peak demand more effectively.

# Real-World Impact

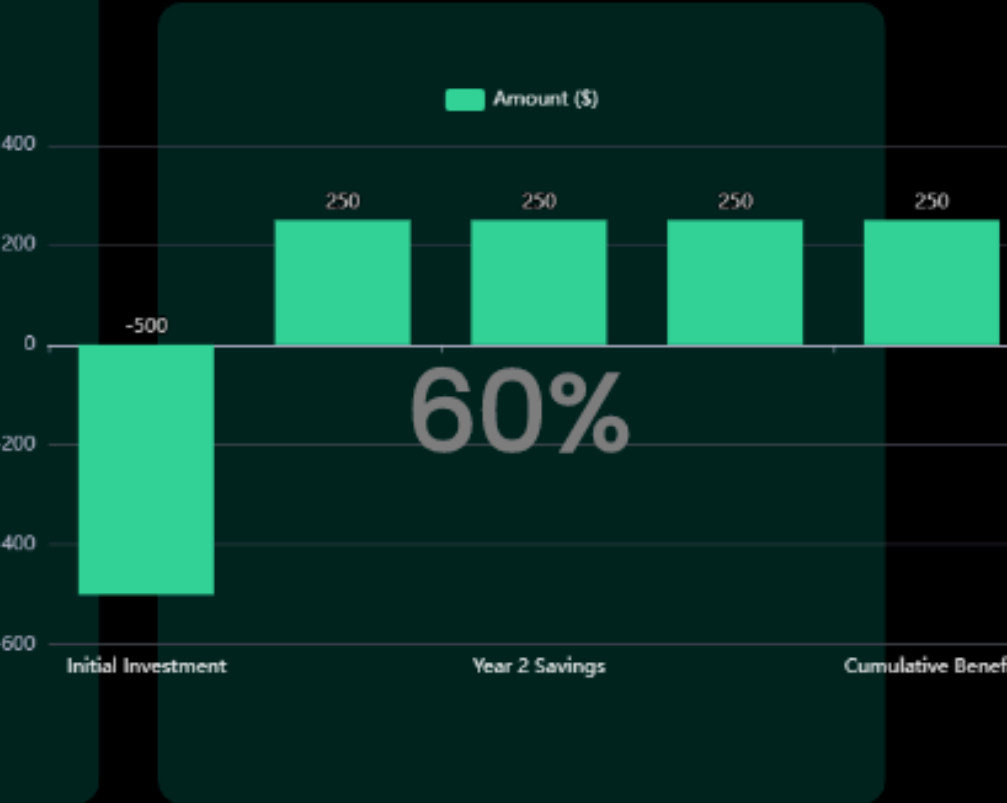
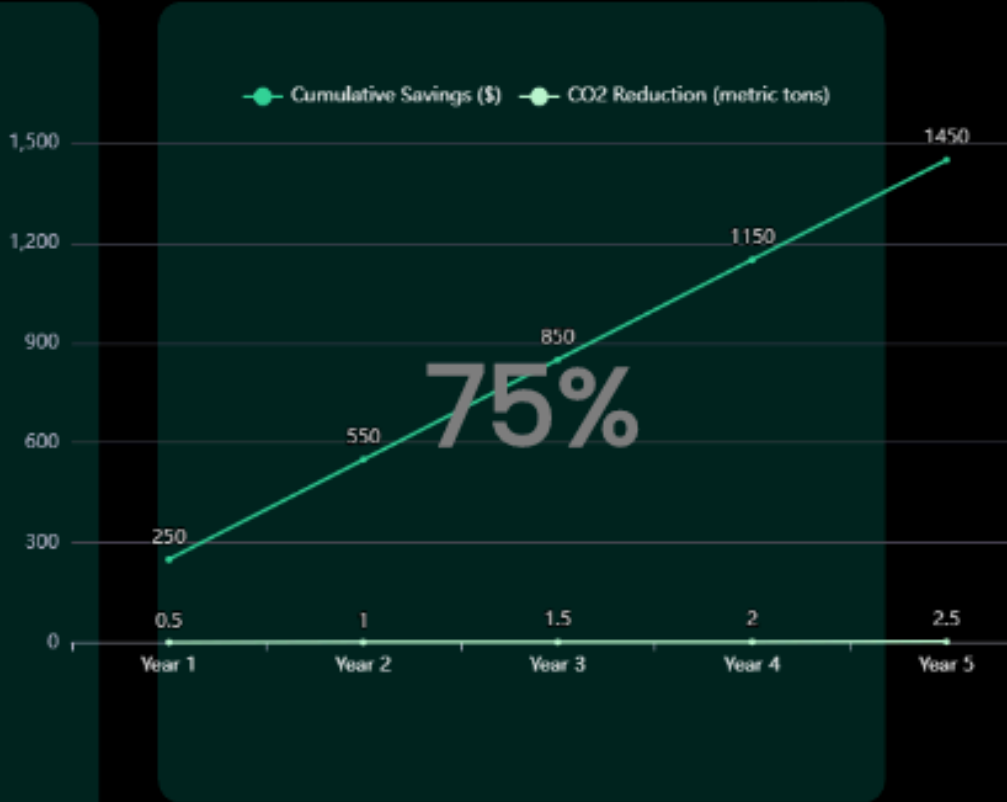
| Metric            | Personal Impact  | Scaled Impact (1M Households) |
|-------------------|------------------|-------------------------------|
| Annual Savings    | \$200-300        | \$200M-300M                   |
| Energy Reduction  | 15-20%           | Significant Grid Relief       |
| CO2 Reduction     | ~0.5 metric tons | ~500K metric tons             |
| Payback Period    | 2-3 years        | Rapid ROI                     |
| System Investment | \$500            | \$500M Total                  |
| Car Equivalent    | -                | 100K vehicles removed         |

## Financial Benefits

Average households save 200M-300M yearly. Smart home investment of \$500 recovers within 2-3 years through operational efficiencies and reduced energy costs.

## Environmental Impact

Smart home adoption reduces CO2 emissions by approximately 500,000 metric tons annually, equivalent to removing 100,000 vehicles from roads. This significant environmental benefit demonstrates the collective impact of individual household efficiency improvements on global sustainability.



# Future Scope & Next Steps

## Phase 1: Energy Forecasting Model

Develop advanced energy forecasting capabilities using ARIMA, Prophet, or machine learning algorithms to predict hourly consumption patterns. This foundation phase establishes accurate baseline predictions essential for subsequent optimization stages, enabling households to anticipate energy needs and plan consumption accordingly.

ARIMA

Prophet

## Phase 2: Anomaly Detection System

Implement real-time anomaly detection system to identify appliance failures and unusual consumption behavior patterns. This detection phase ensures immediate flagging of irregularities, enabling prompt intervention and maintenance to prevent energy waste and equipment damage.

SVM

Autoencode

LSTM

## Phase 3: Smart Scheduling Algorithm

Create intelligent load shifting and demand response participation algorithms to optimize energy distribution during peak and off-peak hours. This optimization phase maximizes efficiency by automatically scheduling non-critical appliances during lower-cost periods, reducing overall household energy expenditure.

GA

DRP

LPO

## Phase 4 & 5: Dashboard & ML Personalization

Deploy interactive dashboard for consumption visualization and goal tracking alongside machine learning personalization that adapts recommendations to household-specific behavior patterns. Complete implementation within 3-6 months, with full integration enabling users to monitor savings and receive customized insights while supporting collaborative opportunities for sponsorship and partnerships.

ML PE

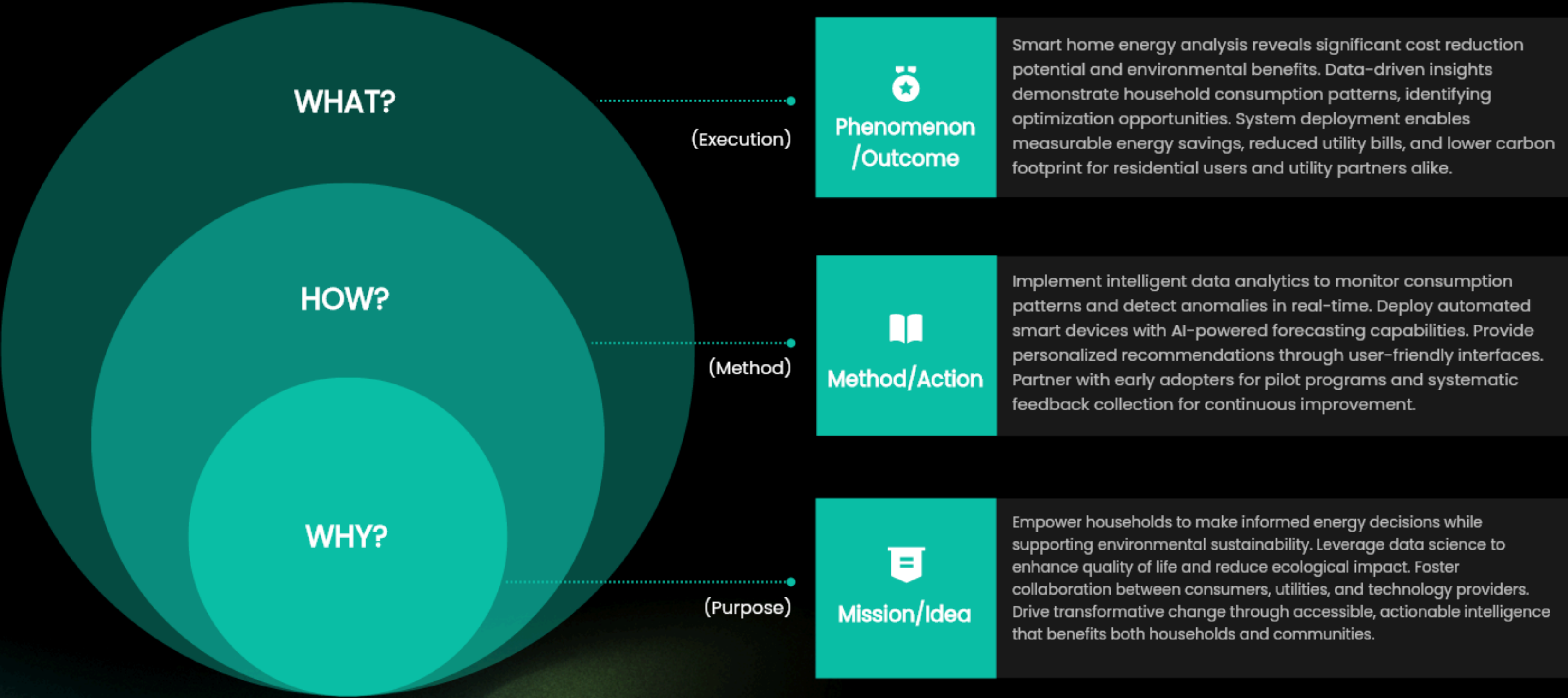
UBPM

RS

Foundation &  
Detection  
Phases

Optimization & Integration  
Phases

# Energy Management Innovation



# Page Count Summary

## Sections 1-2

### Foundation Section

Establishes the presentation framework with four essential slides covering presentation title, problem statement definition, comprehensive dataset introduction, and clear project objectives.

### Analysis Section

Explores data through four detailed slides including data preprocessing techniques, exploratory data analysis findings, identification of top energy consumers, and analysis of consumption peak patterns.

VS

## Sections 3-4

### Insights Section

Delivers four critical slides examining weather impact correlations, outlier detection and analysis, extraction of key insights from data, and actionable recommendations for optimization.

### Conclusion Section

Concludes with three impactful slides demonstrating real-world implementation applications, discussing future research scope possibilities, and synthesizing overall project conclusions.

*Recommendations: Implement smart scheduling  
during off-peak hours • Deploy weather-responsive  
controls • Prioritize high-consumption appliance  
upgrades*

# THANK YOU

